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Often Trusted but Never (Properly) Tested: Evaluating Qualitative Comparative Analysis

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Abstract

To date, hundreds of researchers have employed the method of Qualitative Comparative Analysis (QCA) for the purpose of causal inference. In a recent series of simulation studies, however, several authors have questioned the correctness of QCA in this connection. Some prominent representatives of the method have replied in turn that simulations with artificial data are unsuited for assessing QCA. We take issue with either position in this impasse. On the one hand, we argue that data-driven evaluations of the correctness of a procedure of causal inference require artificial data. On the other hand, we prove all previous attempts in this direction to have been defective. For the first time in the literature on configurational comparative methods, we lay out a set of formal criteria for an adequate evaluation of QCA before implementing a battery of inverse-search trials to test how this method performs in different recovery contexts according to these criteria. While our results indicate that QCA is correct when generating the parsimonious solution type, they also demonstrate that the method is incorrect when generating the conservative and intermediate solution type. In consequence, researchers using QCA for causal inference, particularly in human-sensitive areas such as

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public health and medicine, should immediately discontinue employing the method's conservative and intermediate search strategies.

Keywords

Qualitative Comparative Analysis, method evaluation, set theoretic methods, Boolean algebra, causal inference, configurational comparative methods

Introduction

When judged by its pace of diffusion, Qualitative Comparative Analysis (QCA) is a methodological success story. Since its introduction in the mid-1980s by U.S. sociologist Charles Ragin (1987), the method has spread across the disciplines of business administration, environmental science, international relations, management, political science, public health, and sociology, with a significant body of empirical work having appeared in some of the most respected periodicals of these fields. In other words, QCA has long been trusted by many academic communities to do what researchers have expected it to do: permitting causal inferences on the basis of configurational data by means of Boolean minimization.[1]

Against this backdrop, several authors have recently argued that QCA has, in point of fact, never been thoroughly benchmarked against common standards of evaluation for methods of causal inference. Informed by their own tests, they have cast doubt on QCA, ranging from tentative suggestions to dispense with the algorithms in use to calls for immediate abandonment of the entire approach (Clark, Gilligan, and Golder 2006; Collier 2014b; Krogslund, Choi, and Poertner 2015; Lieberson 2004; Lucas and Szatrowski 2014; Munck 2016; Paine 2016; Seawright 2014; Tanner 2014). For example, while Collier (2014a:124) is convinced that "QCA scholars would do well to [. . .] shift to core building blocks: the methodology of case knowledge and possibly the basic truth table," Lucas and Szatrowski (2014:3) dismiss QCA *in toto* as "a wholly ineffective research method, providing a fatal distraction that, far from realizing the promise of qualitative and comparative historical research, sabotages such analyses."

Many applied users may, therefore, presently be of two minds about the method's status, whereas QCA's severest critics seem to have closed theirs already: Social scientists should "consider turning to other methodological traditions when dealing with the kinds of causal structures with which QCA is intended to work" (Seawright 2014:121). In opposition, we claim that applied users need not turn away from QCA, at least not completely, for two simple reasons. First, we show studies that have disputed the utility of this method to have reported incorrect results or to have employed unsuitable tests; and second, we find that QCA performs faultlessly in a series of methodologically appropriate tests, but only in conjunction with the parsimonious solution type.[2] When producing conservative and intermediate solutions, the latter of which have long been recommended by some methodologists to users of QCA (Ragin 2008:171; Schneider and Wagemann 2012:175), we find the method to be untrustworthy.

However, we also take issue with the out-of-hand rejection of simulations by some prominent representatives of QCA (Olsen 2014; Ragin 2014; Ragin and Rihoux 2004). We argue not only in agreement with Lucas and Szatrowski (2014), but, what is more, also the vast literature on causal inference, that rigorous testing requires *simulated* data (e.g., Hyttinen, Eberhardt, and Hoyer 2012; Kleinberg 2013; Morgan and Winship 2007; Pearl 2009b). The rejection of such data due to the impossibility of gathering empirical case knowledge for doing justice to some researchers' perspective of QCA as an "approach" is self-defeating.

This article is structured as follows. The second section lays out the theoretical foundations of rigorous method evaluation and introduces the design of inverse-search tests that do justice to the standards of QCA. In the third section, we then address two problems that have prevented progress in the evaluation of QCA: the refusal to accept simulated data by some of the method's most prominent representatives and the inadequacy of tests conducted by many critics. In the fourth section, we design four basic tests of varying complexity in order to assess whether QCA is able to recover presupposed causal structures under different conditions. Finally, the concluding section recapitulates our argument, emphasizes some of its major implications, and points toward future avenues of research. So as to ensure full transparency, an annotated replication script for all tests carried out in this article is available in the Online Appendix.

A Theoretical Primer on Method Evaluation

This section lays out the theoretical requirements for data-driven evaluations of causal inference procedures in general and, subsequently, of QCA in particular.

Correctness, Completeness, and Informativeness of a Causal Inference Procedure

The foremost criterion any algorithmic inference procedure has to comply with is *correctness*, that is, its output must conform to its intended purpose or specification (Cormen et al. 2009:6; Dasgupta, Papadimitriou, and Vazirani 2008:3; Hoare 1969:579-80).[3] As the purpose of a procedure of *causal* inference is to infer causal relations underlying the input data, it is correct if, and only if (iff), its causal conclusions are true; or negatively framed, iff it does not commit causal fallacies.

This general characterization calls for three qualifications. First, it cannot be demanded that causal inference procedures be correct independently of the quality of the data they are to process, since true conclusions cannot be systematically inferred from deficient data. Whether data meet stipulated quality standards is determined by whether the causal structure that generates them is adequately reflected. As this structure is unknown and being searched for in empirical work, data quality cannot be established analytically in real-life contexts of causal discovery. Instead, it must be imposed by assumption (Pearl 2009a:99-102). Even heuristics designed to ensure compliance with these assumptions, such as randomization and experimental control, cannot completely eliminate insufficient data quality as a possible source of error. Accordingly, all procedures of causal inference come with a set of background assumptions and can only be expected to produce correct results provided these assumptions are satisfied.[4] A procedure of causal inference must not be blamed for committing a causal fallacy if one or more of its respective assumptions are violated.

Second, empirical data often underdetermine their own causal modeling by giving rise to multiple models that fit these data equally, despite the fulfillment of all quality standards. Procedures in every methodological framework are regularly confronted with such situations of underdetermination in which data do not reflect their generating structure unambiguously (Eberhardt 2013; Simon 1954; Spirtes, Glymour, and Scheines 2000:59-72). As no procedure can be expected to disambiguate what is empirically underdetermined, all data-fitting models must be brought to the attention of the analyst. Not all of these models may correctly reflect the underlying causal structure, but it must be guaranteed that the latter is truthfully reflected by at least one presented model (Spirtes et al. 2000:81). If a procedure outputs, say, three models $\mathbf{m}_i$, $\mathbf{m}_j$, and $\mathbf{m}_k$, it determines that the data-generating causal structure has the form of $\mathbf{m}_i$ *or* that of $\mathbf{m}_j$ *or* that of $\mathbf{m}_k$, a disjunction which is true iff at least one disjunct is true.

Third, causal structures may have a wide range of qualitative and quantitative properties, including functional and spatiotemporal architectures, or the strengths of causal bonds. However, no currently available procedure of causal inference is capable of revealing all of these properties. A correct procedure thus need not uncover all properties of a data-generating structure, but it must uncover those properties which it is designed to identify.

Against the background of these qualifications, the *correctness* of a causal inference procedure can now be defined more precisely: A procedure of causal inference $\mathscr{P}$ is correct iff, whenever $\mathscr{P}$ identifies a set of models $\mathbf{M}$ that fit the analyzed data δ, which comply with $\mathscr{P}$'s background assumptions $\mathbf{B}(\mathscr{P})$ equally well, then the causal properties represented by at least one model $\mathbf{m}_i \in \mathbf{M}$ are a subset of those properties of the δ-generating causal structure Δ which $\mathscr{P}$ is designed to uncover.

This criterion has two important implications. On the one hand, $\mathscr{P}$ need not identify a set of causal models from every set of data. Empirical data may be insufficient to warrant any causal inference. Whenever $\mathscr{P}$ abstains from such an inference, it cannot commit a causal fallacy. By extension, correctness cannot be violated. On the other hand, it may be impossible for $\mathscr{P}$ to uncover all properties of Δ from δ as real-life data tend to be fragmentary. Correctness merely demands that, if $\mathscr{P}$ outputs a set $\mathbf{M}$, then at least one model $\mathbf{m}_i \in \mathbf{M}$ be such that all causal properties represented by $\mathbf{m}_i$ truthfully reflect *some* properties of Δ.

Yet, if $\mathscr{P}$ is given data that are not fragmented, it is fair to require that $\mathscr{P}$ uncovers all properties of Δ. *Completeness* is therefore imposed as a conditional criterion: If $\mathscr{P}$ is given nonfragmented data in compliance with $\mathbf{B}(\mathscr{P})$, $\mathscr{P}$ is complete iff the properties represented by at least one model $\mathbf{m}_i \in \mathbf{M}$ truthfully reflect *all* properties of Δ.

Because of the fact that data fragmentation is ubiquitous in observational studies, however, procedures employed in this domain usually cannot be expected to be complete. From fragmented data, $\mathscr{P}$ should merely be required to uncover all those properties for which δ contains evidence, not fewer and not more. More specifically, $\mathscr{P}$ must be *informative* in the following sense: All and only those properties of Δ for which δ contains evidence are truthfully reflected by at least one model $\mathbf{m}_i \in \mathbf{M}$ (cf. Spirtes et al. 2000:289).

Configurational Correctness

Before adequate test designs can be discussed, correctness as the core quality benchmark of causal inference procedures must be adapted to the particularities of QCA. Most of all, to be able to decide whether a model $\mathbf{m}_i$ truthfully reflects those properties of a causal structure Δ, which the method claims to be able to uncover, the nature of these properties and the interpretation of QCA models must be clarified.

QCA analyzes a set of configurational data δ that has the form of $n \times k$ matrices, where n is the number of units of observation (cases) and k is the number of factors in an analyzed factor frame **F**. **F** must be divided into a first subset comprising exactly one endogenous factor, from which the outcome is selected, and a second subset comprising all remaining factors as exogenous factors. Based on δ, QCA identifies causally relevant levels of the exogenous factors in **F** via an intermediary device known as a *truth table* and a process of rigorous *Boolean minimization*. Subsequently, it groups them into conjunctive and disjunctive causes of the outcome. For instance, a typical causal model returned by QCA is model $\mathbf{m}_a$ in expression (1):

$$\mathbf{m}_a : \neg X_1 X_2 \ \vee \ X_2 \neg X_3 \ \vee \ X_4 \ \Leftrightarrow \ Y. \tag{1}$$

From a purely functional perspective, $\mathbf{m}_a$ claims that the negation of X_1 in conjunction with X_2, X_2 in conjunction with the negation of X_3, and X_4 are three alternative, minimally sufficient conditions of Y whose disjunction is a minimally necessary condition of Y. Against the background of Mackie's (1974) theory of causation implemented by QCA, these functional dependencies can be interpreted in terms of the following causal claim: levels $\neg X_1, X_2, \neg X_3$, and X_4 of their respective factors are each causally relevant for Y, and $\neg X_1$ and X_2 are located on the same causal path to Y, which differs from the path on which X_2 and $\neg X_3$ are located, which again differs from the path on which X_4 is located. In other words, QCA models ascribe causal relevance to their constitutive factor levels and place them on the same or different paths to the outcome.

Two fundamentals of the interpretation of QCA models should be reiterated. First, QCA models exclusively make claims about causal relevance, not about causal irrelevance. According to QCA's background theory of causation, X_i is a cause of an outcome Y iff there exists a configuration $X_j X_k \ldots X_n$ in combination with which X_i makes a difference to Y (Mackie 1974:59-87). While establishing causal relevance merely requires demonstrating the existence of at least one such difference-making scenario, establishing causal irrelevance would require demonstrating the nonexistence of such a scenario, which is impossible on the basis of nonexhaustive data. Correspondingly, the fact that, say, a further factor level X_5 does not figure in model $\mathbf{m}_a$ does not imply X_5 to be causally irrelevant to Y. The noninclusion of X_5 simply means that the data from which model $\mathbf{m}_a$ has been derived do not contain evidence for the causal relevance of X_5 with regard to Y. Future research having access to additional data might reveal X_5 to be causally relevant to Y after all.

Second, QCA models are to be interpreted relative to that δ from which they have been derived. They do not purport to reveal all of Δ's Boolean properties but simply detail those causally relevant factor levels along with those conjunctive and disjunctive groupings for which δ contains evidence. By extension, two different QCA models $\mathbf{m}_i$ and $\mathbf{m}_j$ derived from two different data sets δ_i and δ_j are in no disagreement if the causal claims entailed by $\mathbf{m}_i$ and $\mathbf{m}_j$ stand in a subset relation. QCA models are only in competition if they entail causal claims that contradict each other. For example, model $\mathbf{m}_b$ in expression (2) and model $\mathbf{m}_c$ in expression (3) do not conflict with model $\mathbf{m}_a$:

$$\mathbf{m}_b : \neg X_1 \quad \vee \qquad\qquad X_4 \quad \Leftrightarrow \; Y, \tag{2}$$

$$\mathbf{m}_c : \neg X_1 X_2 \; \vee \; X_2 \neg X_3 \; \vee \; X_4 X_5 \Leftrightarrow \; Y. \tag{3}$$

In harmony with model $\mathbf{m}_a$, $\mathbf{m}_b$ assigns causal relevance to $\neg X_1$ and X_4 on different paths. Analogously, model $\mathbf{m}_c$ expresses all that is stated by $\mathbf{m}_a$ and adds X_5 to the path of X_4. The causal claims entailed by $\mathbf{m}_b$ thus constitute a subset of the claims entailed by $\mathbf{m}_a$, which in turn are contained in the claims of $\mathbf{m}_c$. All three models simply describe properties of one and the same Δ at different degrees of detail and relative to different sets of data δ_a, δ_b, and δ_c.

A second prerequisite for an adaptation of the notion of correctness to the particularities of QCA consists in spelling out the background assumptions $\mathbf{B}(\mathscr{P})$ configurational data have to satisfy.[5] Herein, we confine ourselves to one assumption that can be shown to be sufficient for the correctness of QCA and that is easily satisfiable in simulation-based method evaluation.

On equal terms with other methods of data analysis, QCA is affected by what Holland (1986:947) has dubbed the *fundamental problem of causal inference*, according to which the difference a cause makes to its effect can never be examined on the same unit of observation. Difference-making relations must always be observed on similar but different units, whereby the possibility of confounding is inevitably introduced. Generally put, configurational data are confounded iff certain unmeasured causes change between observed cases in such a way that resulting differences in the outcome appear to be due to the measured factors, whereas they are actually due to the changing unmeasured causes. Confounding is most likely to occur if, in Δ, two measured factor levels Y_1 and Y_2 are causally unrelated parallel effects of an unmeasured common cause W. In such a scenario, δ will neither feature $Y_1 \neg Y_2$ nor $\neg Y_1 Y_2$, which entails that Y_1 and Y_2 are mutually sufficient for each other in δ. If one of these factor levels is specified as the outcome of interest, QCA will erroneously ascribe causal relevance to the other. So as to avoid such fallacies, QCA must assume that the behavior of the factors in $\mathbf{F}$ is not confounded.

To spell out the details of this assumption, we first have to identify the subset of an outcome's unmeasured causes that can confound configurational data, for not all of them have this capacity. Configurational methods aim to uncover the causes of an outcome Y within an analyzed factor frame $\mathbf{F}$ by comparing cases that show differences in the levels of Y and tracing these differences back to variations in the levels of exogenous factors in $\mathbf{F}$. Clearly, though, any two cases with different levels on Y differ not only with respect to measured exogenous factors but also with respect to the latter's causal ancestors and descendants, that is, with respect to a multitude of unmeasured causes of Y that are located on causal chains through the elements of $\mathbf{F}$. Yet all unmeasured causes of Y whose complete causal influence on Y is mediated through the elements of $\mathbf{F}$ cannot induce causal fallacies since their variation is indirectly controllable via the measured variation in $\mathbf{F}$. Factors that can induce causal fallacies are unmeasured causes of Y that change the level of *Y independently* of the elements of $\mathbf{F}$. Changes in the levels of Y that are not mediated through $\mathbf{F}$ may be erroneously ascribed to a measured factor that merely happens to covary in its levels with Y without being causally relevant to Y. That is, configurational data δ can be confounded by unmeasured alternative causes of Y or by unmeasured conjuncts of complex causes of Y. More formally: a factor level W is a *configurational confounder* of an outcome Y relative to an analyzed factor frame $\mathbf{F}$ iff $W \notin \mathbf{F}$ is a cause of Y that, in Δ, is located on a causal path to Y that does not contain any of the measured factors in $\mathbf{F}$ apart from Y, or that is a conjunct in a complex cause of Y, some of whose elements are contained in $\mathbf{F}$.

Configurational data δ on a frame $\mathbf{F}$ are not confounded if all confounders of Y relative to $\mathbf{F}$, which constitute the set $\mathbf{W}(Y, \mathbf{F})$, remain constant across all cases in δ. More specifically, an assumption that is sufficient to exclude confounding stipulates that δ are *homogenous* in the following sense:

> Configurational homogeneity (**CH**): Configurational data δ for an outcome Y over a factor frame $\mathbf{F}$ are homogenous iff every confounder in $\mathbf{W}(Y, \mathbf{F})$ is present in all cases in δ or absent in all cases in δ.

Requiring δ to comply with **CH** amounts to a strong assumption that may be difficult to justify in observational studies. However, on a par with background assumptions in other methodological frameworks, the function of **CH** is merely to *guarantee* the correctness of QCA. If data δ are homogenous, it follows that all observed differences in the outcome must be due to variations of the measured factors, which, in turn, ensures that QCA cannot commit fallacies by ascribing the difference-making relations it uncovers in δ to causal influences of the measured factors (cf. Hofmann and Baumgartner 2011). At the same time, it is important to emphasize that, if QCA is applied to data that violate **CH**, it does not follow that incorrect results will be generated.[6] It only follows that correctness is no longer guaranteed if QCA is applied to inhomogenous data. We are now in a position to lay out the details of the conditions under which a configurational method proves correct:

Configurational correctness (**CC**): A configurational method $\mathscr{P}_c$ is a correct procedure of causal inference iff whenever $\mathscr{P}_c$ infers a set of models $\mathbf{M}$ from a set of data δ which complies with **CH**, then (at least) one model $\mathbf{m}_i \in \mathbf{M}$ satisfies the following three conditions:

(1) all levels of exogenous factors contained in $\mathbf{m}_i$ are causally relevant for the corresponding outcome in the δ-generating structure Δ,
(2) if two levels of exogenous factors X_1 and X_2 are contained in two different disjuncts in $\mathbf{m}_i$, then X_1 and X_2 are located on two different causal paths in Δ, and
(3) if two levels of exogenous factors X_1 and X_2 are contained in the same conjunct in $\mathbf{m}_i$, then X_1 and X_2 are part of the same complex cause in Δ.

We refer to the set of all models satisfying **CC**(1) to **CC**(3) for a given Δ as $\mathbf{M}_{\mathbf{CC}}(\Delta)$.

Testing Configurational Correctness

In light of these clarifications of the conditions under which configurational methods can be said to pass as correct, it should come as no surprise that data-driven tests of whether QCA satisfies **CC** must draw on simulated data.[7] In order to ensure that **CH** is satisfied, Δ must be known. In real-life contexts of causal discovery, however, Δ is unknown and compliance with **CH** can only be rendered plausible via heuristics but never ascertained beyond doubt. If QCA fails to find Δ when applied to real-life data, this failure cannot conclusively be ascribed to deficiencies of QCA; it might just as well be due to deficient data. In real-life discovery contexts, any procedure of causal inference can thus always be immunized against causal fallacies.

The standard design of proper correctness tests for procedures of causal inference builds on so-called *inverse searches*, which reverse the order of causal discovery as it is normally conducted in scientific practice. An inverse search comprises three steps: (1) a causal structure Δ that complies with the type of structures targeted by a procedure $\mathscr{P}$ is stipulated, (2) artificial data δ

are generated by letting the involved factors behave in accordance with Δ, and (3) δ is processed by $\mathscr{P}$ with the aim of identifying the relevant properties of Δ. The inverse search is successfully completed iff those properties are recovered (cf. Claassen and Heskes 2010; Diamond and Sekhon 2013; Hyttinen et al. 2012:3410-15; Kleinberg 2013:183-203; Pearl 2009b:49-57; Spirtes et al. 2000:80-91).

Inverse searches allow for rigorous correctness testing because $\mathscr{P}$ cannot be immunized against unsuccessful results. Based on the presupposed Δ, it is possible to simulate *ideal* data—data that are free of all deficiencies of their real-life cousins. Against such an idealized background, a failure to find Δ can be directly blamed on $\mathscr{P}$. Moreover, as all properties of Δ that $\mathscr{P}$ is designed to uncover should be inferable from ideal data, inverse searches also permit testing the completeness of $\mathscr{P}$. In addition, it is possible to simulate nonideal data, which allow for evaluating the informativeness of $\mathscr{P}$'s conclusions and, more generally, for exploring to what degree a gradual deterioration in the quality of data affects the quality of $\mathscr{P}$'s output.

In the fourth section, we investigate QCA's methodological characteristics in a variety of inverse search series, in the course of which we simulate ideal data before gradually introducing different types of data deficiencies. First, however, the third section reviews recent evaluations of QCA that have been based on data simulations.

Problems of Existing Research

Two major problems have prevented progress in the current debate over the utility of QCA. The first, created by some representatives of the method, is the out-of-hand rejection of artificial data. For example, confronted with a simulation by Lieberson (2004), Ragin and Rihoux (2004:23) retort that "both of his hypothetical data sets fall from the sky and are not constructed from case materials [. . .]. For these reasons, the two data sets would be irrelevant to QCA." Similarly, Olsen (2014:102) concludes in her comment on Lucas and Szatrowski (2014) that the argument using simulated data "does not refute QCA, because no background knowledge can be obtained: these data do not reflect reality." These authors reject simulated data because empirical case knowledge, one of the method's putative requirements, cannot be gathered.

The importance of such knowledge has been defended by invoking a distinction between QCA as an "approach" and QCA as a "method," with the latter constituting the bare algorithmic machinery that is embedded in a larger exercise of acquiring intimate case knowledge (Rihoux 2006:683;

Schneider and Wagemann 2012:8-13). Ragin (2014:80-81), for instance, argues that Lucas and Szatrowski (2014) "limit their discussion to QCA as a *method* of data analysis. However, it embraces several key elements that together define it as an *approach*. Any critique that fails to address these interdependent aspects is not a critique of QCA."[8] Clearly, such dismissals are self-defeating. If a method is centrally embedded in the larger context of an approach (Rihoux and Lobe 2009; Schneider and Rohlfing 2013:561), and the former is shown to be faulty with simulated data that permit control over all sources of error, then these deficiencies *must* feed into the latter. A machinery producing incorrect solutions will necessarily lead researchers astray in their ensuing attempts to make sense of results on the case level. If properly designed and adequately implemented simulations demonstrate that QCA does not work correctly, then the method cannot be accepted as a reliable procedure of causal inference.[9]

In this connection, the second problem in the recent discussion on the utility of QCA has originated in the misapplication of simulations. Despite the recognition that "[s]imulations must fit the method being evaluated" (Collier 2014a:123), a lack of sufficient understanding of the algebraic foundations of QCA has been rife (Thiem and Baumgartner 2016a; Thiem, Baumgartner, and Bol 2016). Often, critics have tried to recover regression models, or parts thereof, in consequence of which conclusions were guaranteed to be that "QCA [. . .] fails to find the correct interactions when interactions are present, selects the wrong direction of association, and finds asymmetric causation when the known causal structure is symmetric" (Lucas and Szatrowski 2014:3). That QCA has not been designed to identify regression models, or any parts thereof, for that matter, has been ignored by these critics.[10]

Moreover, when no attempts at recovering regression models with QCA were made, the method was misapplied or inadequate tests were devised. For example, Lucas and Szatrowski (2014:17-23) construct a hypothetical data set, call it δ_{LS} here for simplicity, of 40 cases under the factor frame $\mathbf{F}_1 = \{X_1, X_2, X_3, X_4, Y\}$. Independent Bernoulli trials are performed for the exogenous factors to simulate observations, and the endogenous factor Y is assumed to take on the value 1 if, and only if, $X_1 = 1$ and $X_2 = 1$ or $X_3 = 1$ and $X_4 = 1$. Translated into Boolean algebra, QCA should therefore be expected to recover model $\mathbf{m}(\Delta_{LS})$ in expression (4):

$$\mathbf{m}(\Delta_{LS}) : X_1X_2 \vee X_3X_4 \Leftrightarrow Y. \quad (4)$$

The authors do not notice that their simulated data are peculiar in that they cause QCA to output the same set of models for all three solution types—

conservative, intermediate, and parsimonious.[11] Nonetheless, Lucas and Szatrowski (2014:18-19) discuss these solution types individually and present separate solutions. Worse still, they appear to have failed in operating their chosen software, fs/QCA (Ragin and Davey 2014). The solutions they present for $Y = 1$ are not the ones returned by fs/QCA, which outputs the correct solution: model $\mathbf{m}(\Delta_{LS})$.[12]

Furthermore, Lucas and Szatrowski (2014) devise inadequate validation tests. So as to test QCA's proclivity for including causally irrelevant factors, the authors add a fifth factor Z_1 to δ_{LS} that is pairwise weakly correlated with every other factor in $\mathbf{F}_1$. They then expect QCA to not attribute any causal relevance to Z_1. Such a design, however, ignores that QCA does not search for correlations but for implications (Thiem et al. 2016).[13] Rendering Z_1 directly irrelevant to Y necessitates not pairwise correlational independence between Z_1 and the factors in $\mathbf{F}_1$ but implicational independence between Z_1 and all combinations of exogenous factor levels resulting from $\mathbf{F}_1$. This cannot be accomplished by simply adding Z_1 to δ_{LS}. The reason is that a fifth exogenous factor increases the space of possible minterms from $2^4 = 16$ to $2^5 = 32$, and Z_1 is only independent of all 16 minterms if it is both positively and negatively combined with each of these minterms, whereby an expanded frame $\mathbf{F}_2 = \{X_1, X_2, X_3, X_4, Z_1, Y\}$ is required. Properly simulating the introduction of an independent factor calls for simulating data not on $\mathbf{F}_1$ but on $\mathbf{F}_2$. By not meeting this requirement, the test by Lucas and Szatrowski (2014) produces dependencies that are nothing but design artifacts (Thiem and Baumgartner 2016b).[14]

Lastly, critics have maintained that QCA solutions tend not to be robust (Hug 2013; Krogslund et al. 2015; Lucas and Szatrowski 2014). However, only against the backdrop of an implicit assumption that different QCA models represent incompatible causal structures can it be deemed problematic that, for example, the deletion of the two cases of $X_4 \neg Y$ from δ_{LS} changes QCA's output from model $\mathbf{m}(\Delta_{LS})$ to model $\mathbf{m}(\Delta'_{LS})$ in expression (5):

$$\mathbf{m}(\Delta'_{LS}) : X_1X_2 \vee X_4 \Leftrightarrow Y. \tag{5}$$

Yet, as set out above in the section on Configurational Correctness, two QCA models whose causal claims stand in a subset relation are perfectly compatible.[15] Model $\mathbf{m}(\Delta'_{LS})$ states that X_1X_2 and X_4 are causally relevant and located on alternative paths to Y, but it does not claim that X_3 is of no causal relevance. The data that result from the elimination of the two cases of

$X_4 \neg Y$ from δ_{LS} simply remove all grounds for establishing the causal relevance of X_3, and QCA therefore abstains from integrating X_3 into $\mathbf{m}(\Delta'_{LS})$. In contrast, $\mathbf{m}(\Delta_{LS})$ states that X_3 is causally relevant and locates this condition on the same path as X_4. Model $\mathbf{m}(\Delta_{LS})$ thus merely makes a stronger causal claim than $\mathbf{m}(\Delta'_{LS})$, but these two claims are entirely compatible; one logically entails the other.[16]

To summarize, QCA is designed to compare configurations and to infer causal dependencies from difference-making scenarios represented in truth tables. That it reacts to changes in the latter the way it does is not a weakness, but an indication that the method works as intended. As long as **CC** is satisfied, varying models inferred by QCA from varying truth tables simply reflect variation in the empirical evidence about the data-generating structure contained in these truth tables. In the next section, we prosecute our inquiry by carrying out a series of tests for evaluating QCA in full accordance with the method's objectives and standards.

Evaluating QCA

We implement all tests using the R package QCApro, which contains extensive evaluation functions for QCA (Thiem 2016b).[17] In addition, QCApro is capable of returning the full model space of a solution, unlike other programs for configurational data analysis (Baumgartner and Thiem 2015b; Thiem 2014b; Thiem and Duşa 2013a).[18]

All our tests adhere to the inverse-search canon laid out in the second section, with one important exception: not data sets will be simulated but truth tables. The reason is that QCA transforms raw data into a truth table in its first algorithmic phase (Thiem 2016a; Thiem, Spöhel, and Duşa 2016:108). One and the same truth table can represent the dependencies among exogenous and endogenous factors in an infinite number of different data sets, whereas the reverse is not true. The direct simulation of truth tables therefore has the major advantage of covering, without generating any redundant information, an unlimited number of data sets, for all of which our conclusions remain valid. Moreover, this strategy frees us from having to impose restrictions on the probability mass function describing the likelihood of observing any particular combination of case attributes. In preparation, we thus construct a truth table $\mathbf{T}_{\Delta^*}$ listing the complete distribution of truth values under the main operator ($\wedge$) of the causal model $\mathbf{m}(\Delta^*)$ in expression (6) that represents the hyperstructure Δ^* underlying all our ensuing tests:

$$\mathbf{m}(\Delta^*) : \underbrace{\overbrace{\overbrace{(D \vee \underbrace{\neg AB \vee B\neg C}_{\text{Test IIb: Underspecification}} \Leftrightarrow Z)}^{\text{Test I: Ideal context}}}^{\text{Test III: Limited empirical diversity}} \wedge (E \vee \neg E)}_{\text{Test IIa: Overspecification}}. \tag{6}$$

Subsequently, the minterms of the exogenous factors A, B, C, D, E, and Z for which $\mathbf{m}(\Delta^*)$ is true are culled from $\mathbf{T}_{\Delta^*}$ to yield the truth table $\mathbf{T}^C_{\Delta^*}$ of $\mathbf{m}(\Delta^*)$-compatible minterms. $\mathbf{T}^C_{\Delta^*}$ subsumes all truth tables analyzed in the remainder of this section. Each test is marked by curly brackets around its corresponding part in expression (6). In the first test, an ideal recovery context is ensured in that the analyzed truth table is saturated and the factor frame is correctly specified.[19] In the second test, complexity is increased by allowing for misspecification of the factor frame insofar as, first, a factor that is causally irrelevant to the outcome is included and, second, a causally relevant factor is omitted. The effects of limited empirical diversity are examined in the third test, where QCA's three solution types are evaluated. We focus on these four particular issues in the present article, because they have been central in recent assessments of QCA. At the same time, it is important to emphasize that our evaluations neither address combinations of the above test designs nor other issues of configurational data analysis, such as inconsistencies, for example.

Test I: Saturated Truth Table, Correct Factor Frame

Lucas and Szatrowski (2014:68-69) argue that "one fundamental issue concerns how a method works under conditions proponents deem ideal," but they conclude "that QCA fails under such conditions." On the one hand, several commentators have shown that incorrect results had informed this assessment (Fiss, Marx, and Rihoux 2014:97; Ragin 2014:92-93; Vaisey 2014:108). On the other hand, however, it is to the credit of Lucas and Szatrowski (2014) that they have been the first to have tried to evaluate the method with simulated data, an indispensable prerequisite for all procedures of causal inference before applied researchers can employ them unreservedly (Diamond and Sekhon 2013; Kleinberg 2013; Morgan and Winship 2007; Pearl 2009b; Spirtes et al. 2000). As Lucas (2014:143) rightly notes in this regard, "[n]o one has a burden of proving QCA does not work; QCA proponents have the burden of proving that QCA does work [. . .]."[20]

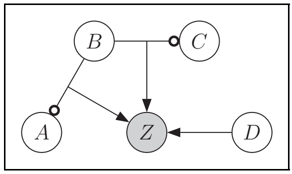

Figure 1. Causal model representing Δ', $\mathbf{m}(\Delta') : \neg AB \vee B\neg C \vee D \Leftrightarrow Z$.

As Lucas and Szatrowski's failed attempt at verifying whether QCA works correctly when the context is ideal has been the first and only one to date, we perform a methodologically clean test anew. The causal structure Δ' we presuppose to this end is visually represented by $\mathbf{m}(\Delta')$ in Figure 1. Each causally relevant condition is represented by a node enclosing the label of its factor, a conjunction of two exogenous factor levels is represented by an undirected edge, a causal dependency between a condition and an outcome by a directed edge from the former to the latter, and the negation of a causally relevant condition by a circle at the respective end of the edge to another condition or at the tail of a directed edge to the outcome. Verbally put, the outcome, Z, results iff at least one of the three presupposed causal paths, $\neg AB$, $B\neg C$, and D, is instantiated.

Having specified the underlying causal structure, the context parameters under which QCA is to be tested need to be delineated. We refer to the recovery context as *ideal* if (1) the truth table to be analyzed is saturated, (2) strictly partial consistencies are absent, and (3) the factor frame is correctly specified. So as to meet criterion (1), empirical diversity must not be limited; to meet criterion (2), the possibility for any two cases to exhibit the same minterm yet a different outcome must be excluded; and to meet criterion (3), those, and only those, factors that are directly relevant to Z in Δ' must be included in the factor frame $\mathbf{F}_{\Delta'}$.

To simulate such a recovery context, we cull $\mathbf{T}^{C}_{\Delta'}$ from $\mathbf{T}^{C}_{\Delta^*}$ by eliminating the column of the irrelevant factor E and by removing duplicate rows. Table 1, which reports the analysis of $\mathbf{T}^{C}_{\Delta'}$ by QCA, shows that Δ' has been completely recovered in $\mathbf{m}_{\mathrm{I}}$, with the corresponding consistency and coverage statistics in column Con (consistency), Cov_r (raw coverage), and Cov_u (unique coverage). These statistics of course only refer to that single data set $\delta_{\Delta'}$ needed to instantiate $\mathbf{T}^{C}_{\Delta'}$ in the software, for which each structurally compatible minterm contains a single case, but any data set resulting from an n-fold duplication of this minimal instance would return exactly the same figures.

All three alternative causes of Z show fully accurate consistency and coverage scores. The raw coverage of D must be twice as high as that of the

Table 1. Solution and Prime Implicant Details for Test I.

Outcome	Prime Implicant	Con	Cov_r	Cov_u
Z	D	1.000	0.727	.455
	$\neg AB$	1.000	0.364	.091
	$B\neg C$	1.000	0.364	.091
	$\mathbf{m}_1$	1.000	1.000	

other two paths since it covers twice as many minterms. Its unique coverage must be five times as high since it covers five minterms that are not covered by any of the other two paths ($DABC$, $DA\neg BC$, $DA\neg B\neg C$, $D\neg A\neg BC$, and $D\neg A\neg B\neg C$), while $\neg AB$ and $B\neg C$ each cover only a single case that is not covered by any of the other two paths ($\neg D\neg ABC$ and $\neg DAB\neg C$, respectively).

To summarize the main result of the first inverse-search test under ideal context parameters: QCA has successfully recovered all relevant properties of the presupposed data-generating structure. No grounds exist for doubting the method's correctness, completeness or informativeness at this stage.

Tests IIa and IIb: Saturated Truth Table, Incorrect Factor Frame

QCA has performed faultlessly under ideal context parameters. However, the assumption of no misspecification of $\mathbf{F}_{\Delta'}$ may have set too low a benchmark. Irrespective of the confidence researchers have in the causal relevance of a particular set of conditions, they can ultimately *never* be sure. We thus up the ante. Again assuming the underlying causal structure to be Δ', we now test the properties of QCA in a context of an incorrectly specified factor frame. This problem has two facets: first, *overspecification* occurs when a factor that is causally irrelevant to Z is included in $\mathbf{F}_{\Delta'}$; and second, *underspecification* refers to the situation in which a factor that is directly causally relevant to Z is omitted from $\mathbf{F}_{\Delta'}$. The first case is depicted in panel (a) of Figure 2, the second in panel (b). The absence of a directed edge from E to Z signals the former's causal irrelevance to the latter, while the interrupted, directed edge indicates that D, albeit of direct causal relevance to Z, has been omitted.

For techniques of regression analysis, overspecification is usually no cause for worry as far as bias in estimators is concerned (Gelman and Hill 2007:169-70), but underspecification is only unproblematic as long as an excluded variable is uncorrelated with the regressand, or with any included,

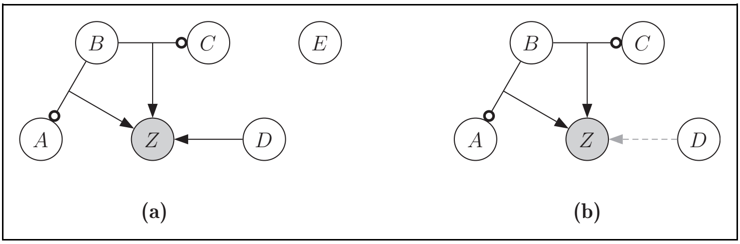

Figure 2. Two forms of factor frame misspecification with respect to $\mathbf{F}_{\Delta'}$. (a) Overspecification: E included in $\mathbf{F}_{\Delta'}$. (b) Underspecification: D omitted from $\mathbf{F}_{\Delta'}$.

Table 2. Solution and Prime Implicant Details for Tests IIa and IIb.

		Overspecification			Underspecification		
Outcome	PI	Con	Cov_r	Cov_u	Inc	Cov_r	Cov_u
Z	D	1.000	0.727	.455			
	$\neg AB$	1.000	0.364	.091	1.000	.364	.182
	$B\neg C$	1.000	0.364	.091	1.000	.364	.182
	$\mathbf{m}_{\text{IIa}}$	1.000	1.000				
	$\mathbf{m}_{\text{IIb}}$				1.000	.545	

Note: PI = Prime Implicant.

causally relevant regressor. One may be tempted to presume that a similar logic carries over to QCA, but critics of the method have argued otherwise. As to overspecification, Lucas and Szatrowski (2014) and Krogslund et al. (2015) assert that QCA is beset by an in-built confirmation bias because it is liable to identify factors as causally relevant which are only weakly correlated or even uncorrelated with the outcome factor. As regards underspecification, Seawright (2005:17) argues that "[i]n order for a Boolean-algebraic analysis to produce useful causal inferences, no variables of any kind that have any causal influence whatsoever on the outcome can be omitted." If these statements held true, QCA should not be expected to eliminate E, and it should commit causal fallacies once D is excluded from $\mathbf{F}_{\Delta'}$. We test these predictions in the remainder of this subsection.

The result of the minimization on the basis of the overspecified factor frame is presented in Table 2. Not only has the irrelevant factor E not survived the minimization process but also has QCA completely recovered Δ' in $\mathbf{m}_{\text{IIa}}$. This outcome is no consequence of happenstance. Recall that we do not model data but truth tables, to which an unlimited number of particular data sets correspond. Through appending the tautology $(E \vee \neg E)$ by conjunction in $\mathbf{m}(\Delta')$, the implicational independence of Z with respect to E is properly operationalized since every minterm conforming to Δ' is combined with E and $\neg E$ (cf. Thiem and Baumgartner 2016b:349-50).

Yet, does underspecification lead to useless models returned by QCA as argued by Seawright (2005)? The results of the minimization on the basis of a specification that omits D from $\mathbf{F}_{\Delta'}$ are presented in the corresponding part of Table 2. Despite the omission of D, which leads to the violation of **CH**, model $\mathbf{m}_{\mathrm{IIb}}$ still truthfully reflects the two other, complex causes. The functional relation reduces from an equivalence to an implication and coverage drops to 0.545, but since all unique cases of DZ cannot be accounted for when D is excluded, a correct method of configurational data analysis should be expected to adjust its solution statistics in this manner.

To summarize the main results of our inverse-search tests in the context of factor frame misspecification: QCA has successfully and completely recovered the presupposed causal structure by correctly eliminating the causally irrelevant factor, and it has successfully recovered the largest part of the presupposed causal structure for which the truth table supplied evidence, despite the omission of a causally relevant factor and the violation of **CH**. Again, no grounds exist for doubting QCA's correctness at this stage. Moreover, our results corroborate the method's informativeness, that is, its capacity to infer all and only those causal properties from data for which the latter contain empirical evidence.

Test III: Unsaturated Truth Table, Correct Factor Frame

Seawright (2014:121) concludes from his simulation that "QCA seems to fail quite often, even under the most favorable conditions, if there is limited diversity." We revisit this conjecture by subjecting each of QCA's three search strategies to all possible degrees of limited diversity.[21] The conservative search strategy of QCA produces the conservative solution type (QCA-CS), the intermediate search strategy the intermediate solution type (QCA-IS), and the parsimonious search strategy the parsimonious solution type (QCA-PS). In the literature, parsimonious solutions are usually regarded with more suspicion than conservative ones but both are considered inferior to intermediate solutions (Ragin 2008:171; Schneider and Wagemann 2012:175).[22] As QCA-IS, unlike QCA-CS and QCA-PS, can generate a range of different solutions for a single data set, we additionally distinguish between QCA-IS_1, in which all directional expectations regarding each factor level are as much in agreement with Δ' as possible, and QCA-IS_2, in which all directional expectations are as much in disagreement with Δ' as possible.[23]

In total, we perform 15 series of test runs, one for each set of minterms of a certain cardinality that can be eliminated from $\mathbf{T}^{C}_{\Delta'}$. The first series comprises $\binom{16}{1} = 16$ runs, one for each of the 16 minterms that can be eliminated; the second series comprises $\binom{16}{2} = 120$ runs, one for each pair of minterms; the third series $\binom{16}{3} = 560$ runs, one for each triple; and so on. The share of runs that have passed the test within a series then yields this series' correctness ratio for a search strategy. Over diversity index values ranging from the (nontrivial) maximum of 93.75 percent (one minterm eliminated) to the (nontrivial) minimum of 6.25 percent (15 minterms eliminated), the number of runs therefore amounts to $4\sum_{i=1}^{15}\binom{16}{i} = 4(2^{16} - 2) = 262,136$.[24] These more than a quarter-million runs cover all possible truth tables that can result on the basis of four exogenous factors.

By the requirements of $\mathbf{CC}(1)$ to $\mathbf{CC}(3)$, a run counts as passed if, and only if, at least one of the models depicted in Figure 3 from $\mathbf{m}_1$ to $\mathbf{m}_{24}$ is returned within any given solution. These 24 models constitute the set of all redundancy-free submodels $\mathbf{M}_{\mathbf{CC}}(\Delta')$ for which correctness remains inviolate. For example, model $\mathbf{m}_{14}$ in panel (n) of Figure 3 preserves correctness because it claims that $\neg A$ and B are causally relevant and connected conjunctively on one causal path to Z, while $\neg C$ is causally relevant on another causal path to Z. These claims fully accord with those made by $\mathbf{m}(\Delta')$. In contrast, a submodel such as $\neg A \neg C \Leftrightarrow Z$ violates $\mathbf{CC}(2)$ because neither conjunct is located on the other conjunct's causal path in $\mathbf{m}(\Delta')$. If no element of $\mathbf{M}_{\mathbf{CC}}(\Delta')$ is returned as part of a solution, the run thus counts as failed.[25] The two limiting cases of $\mathbf{M}_{\mathbf{CC}}(\Delta')$ are model $\mathbf{m}_{23}$ in panel (w) of Figure 3, which represents the true causal structure Δ', and the empty set in model $\mathbf{m}_{24}$ in panel (x). The latter occurs whenever no case of Z or no case of $\neg Z$ survives elimination.

All correctness ratios for model $\mathbf{m}(\Delta')$ (in percentage) along the entire range of diversity values, from 100 percent down to 0 percent, are presented in Figure 4. The line with circles plots the test results of QCA-PS, the line with squares those of QCA-IS$_1$, the line with diamonds those of QCA-IS$_2$ and the line with triangles plots the test results of QCA-CS.

Most strikingly, QCA-PS never generates incorrect solutions; at least one of the models of $\mathbf{M}_{\mathbf{CC}}(\Delta')$ is always present. By comparison, QCA-CS fares much worse. Particularly between diversity values of 81.25 percent and 12.5 percent—the span covering the majority of configurational research

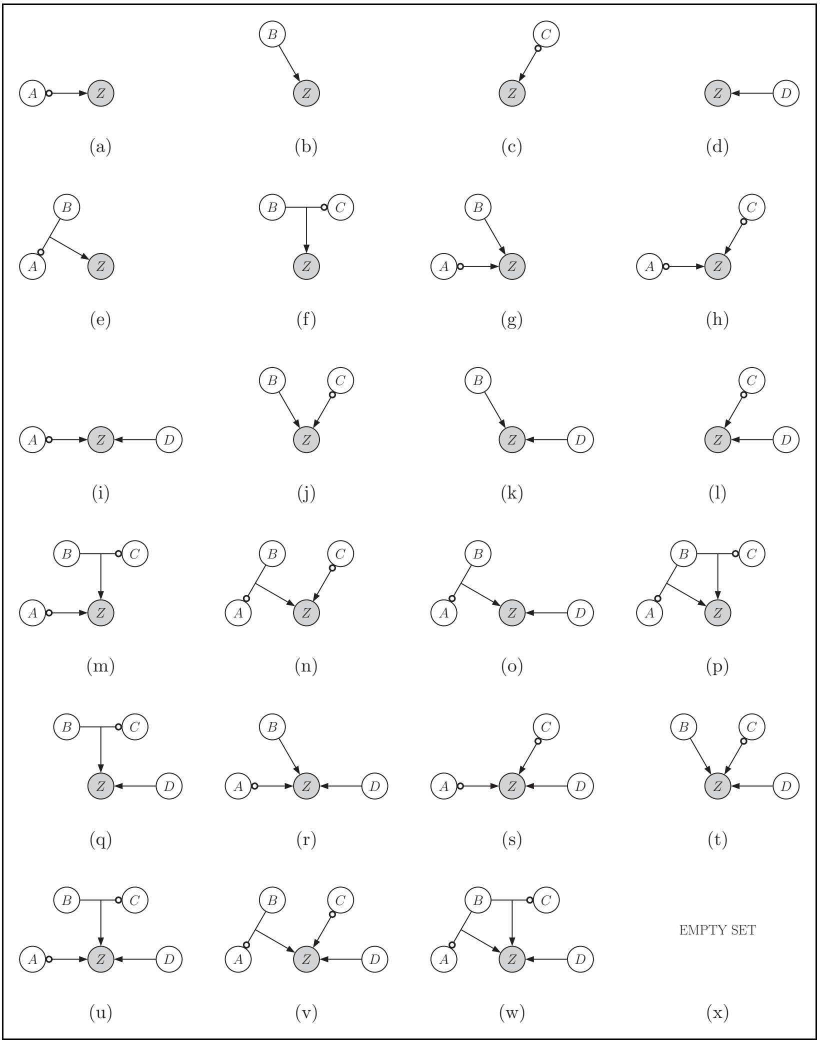

Figure 3. Set of all models $\mathbf{M_{CC}}(\Delta')$ preserving correctness. (a) $\mathbf{m}_1$, (b) $\mathbf{m}_2$, (c) $\mathbf{m}_3$, (d) $\mathbf{m}_4$, (e) $\mathbf{m}_5$, (f) $\mathbf{m}_6$, (g) $\mathbf{m}_7$, (h) $\mathbf{m}_8$, (i) $\mathbf{m}_9$, (j) $\mathbf{m}_{10}$, (k) $\mathbf{m}_{11}$, (l) $\mathbf{m}_{12}$, (m) $\mathbf{m}_{13}$, (n) $\mathbf{m}_{14}$, (o) $\mathbf{m}_{15}$, (p) $\mathbf{m}_{16}$, (q) $\mathbf{m}_{17}$, (r) $\mathbf{m}_{18}$, (s) $\mathbf{m}_{19}$, (t) $\mathbf{m}_{20}$, (u) $\mathbf{m}_{21}$, (v) $\mathbf{m}_{22}$, (w) $\mathbf{m}_{23} = \mathbf{m}(\Delta')$, and (x) $\mathbf{m}_{24}$.

designs—correctness does not even exceed 10 percent.[26] It reaches almost 44 percent at 93.75 percent diversity, and about 31 percent at 6.25 percent

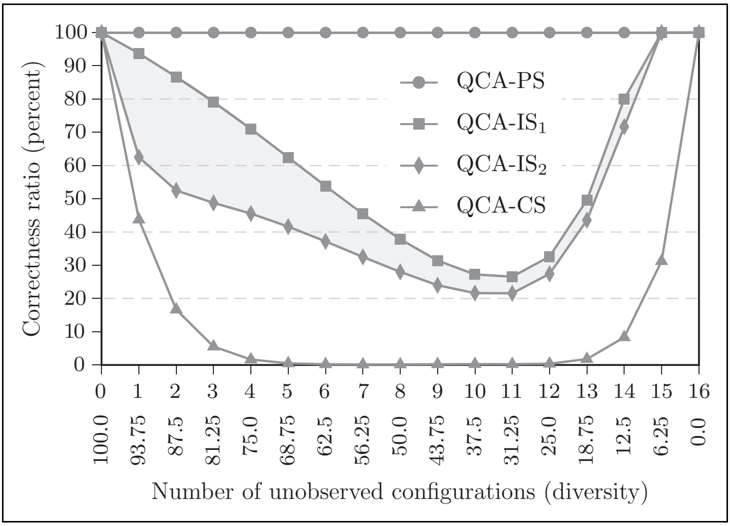

Figure 4. Series correctness ratios for $\mathbf{m}(\Delta') : \neg AB \vee B \neg C \vee D$; submodels: $|\mathbf{M_{CC}}(\Delta')| = 24$.

diversity, but these scenarios, and all closely neighboring ones on the extremer side of either value, are rare cases in empirical applications because they imply almost saturated truth tables, the virtual absence of limited diversity relative to the property space, respectively.

Both QCA-IS variants fall in between the figures for QCA-CS and QCA-PS. Along diversity values down to 62.5 percent, the performance of QCA-IS_1 is closer to that of QCA-PS but decreases linearly at a rate of about 13 percentage points on average, whereas its performance at moderately low diversity values between 56.25 percent and 18.75 percent is closer to that of QCA-CS. QCA-IS_2 performs worse than QCA-IS_1, as would be expected, but the corridor becomes considerably narrower toward lower diversity values.

Model $\mathbf{m}(\Delta')$ was a deliberate choice of convenience because its low complexity implies a limited set of correctness-preserving submodels, which we could easily visualize in Figure 3. However, we may be accused of having also picked this model because it ensures, for whatever reason, the correctness of QCA-PS, and possibly the incorrectness of QCA-IS and QCA-CS. So as to remove these concerns, we perform a demanding robustness test with six additional models $\mathbf{m}(\Delta_1)$ to $\mathbf{m}(\Delta_6)$, each of whose structure is beyond our control. The selection of these six models involves two steps. First, we randomly select six different truth tables from the set of all $2^{16} - 2 = 65,534$ five-factor, nontrivial truth tables; and second, from among all models resulting for each of these truth tables, we again select one model randomly.[27]

The results of this robustness test are presented in panels (a) to (f) of Figure 5.[28] For QCA-PS, the test result remains the same as before with respect to $\mathbf{m}(\Delta')$: It is the only search strategy that passes the test for correctness.[29] Although their respective performance bands are extremely wide, none of the other search strategies ever passes a complete test series. Moreover, yet no less disturbingly from the perspective of current practice to recommend QCA-IS as the most attractive option, solutions for which expectations are less in accordance with the data-generating structure than others sometimes perform better, as in the case of the model in panel (c) of Figure 5 over diversity ranges from 31.25 percent down to 6.25 percent, or that in panel (e) from 18.75 percent down to 6.25 percent, or that in panel (f) from 50 percent down to 25 percent.

To summarize the findings of this subsection: The parsimonious search strategy of QCA recovered at least one correct model under each possible scenario of limited diversity. In contrast, the conservative and the intermediate strategies, the latter in either variant, did not prove correct. At times, the approximation of their respective solution types to the data-generating structure was better than that of the parsimonious search strategy, but not systematically so, and only at the price of violating the primary criterion of configurational correctness.

Conclusions

On the basis of results from a series of simulation studies, QCA has recently been exposed to considerable criticism, ranging from suggestions to dispense with its algorithms to calls for immediate abandonment of the entire approach. In this article, we have shown these critiques to have lacked conclusive evidence that would support taking any of these courses of action. Results proved incorrect, tests were not designed in accordance with the search target of QCA, or test protocols exhibited serious deficiencies.

At the same time, we have acknowledged, in concurrence with some of the severest critics, the long-overdue need for a thorough evaluation of QCA. As such efforts ought to have been initiated by the first generation of the

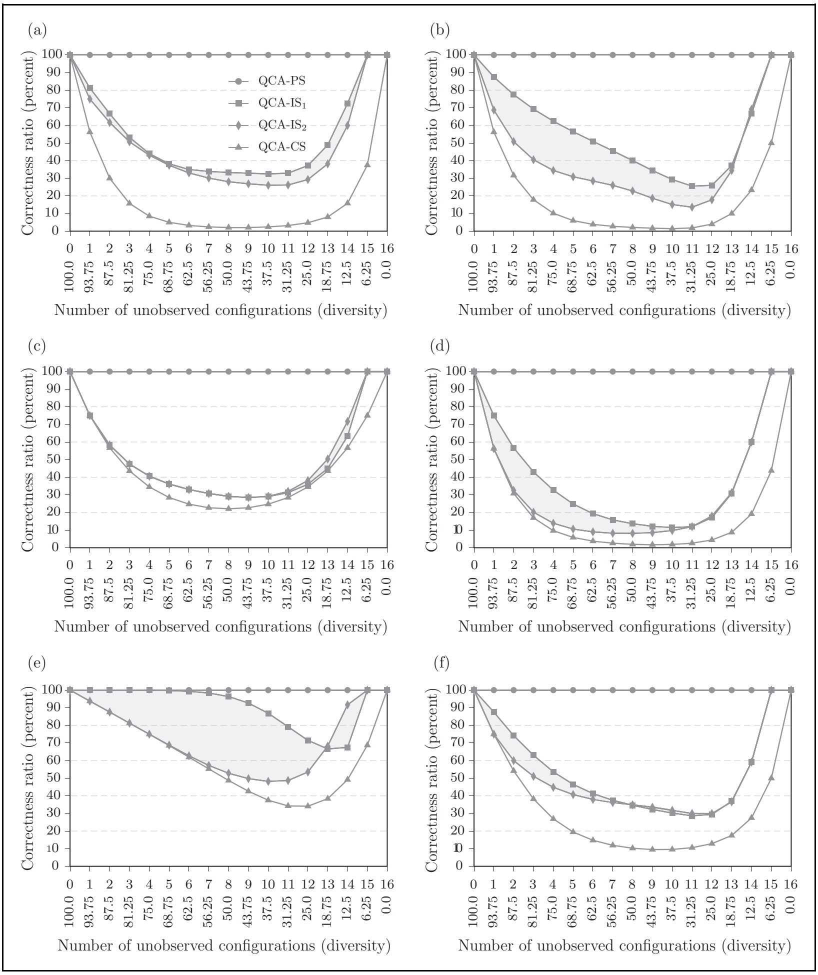

Figure 5. Series correctness ratios for Qualitative Comparative Analysis's three solution types. (a) $\mathbf{m}(\Delta_1) : \neg A\neg D \vee \neg ABC \vee AB\neg C \vee ABD \vee \neg B\neg CD$; submodels : $|\mathrm{M_{CC}}(\Delta_1)| = 705$. (b) $\mathbf{m}(\Delta_2) : \neg A\neg B \vee \neg BC \vee CD$; submodels : $|\mathrm{M_{CC}}(\Delta_2)| = 33$. (c) $\mathbf{m}(\Delta_3) : A\neg CD \vee BCD \vee \neg A\neg B\neg C\neg D$; submodels : $|\mathrm{M_{CC}}(\Delta_3)| = 366$. (d) $\mathbf{m}(\Delta_4) : \neg AD \vee BD \vee \neg A\neg B\neg C \vee AC\neg D$; submodels : $|\mathrm{M_{CC}}(\Delta_4)| = 222$. (e) $\mathbf{m}(\Delta_5) : A\neg B\neg C \vee A\neg B\neg D \vee A\neg C\neg D \vee \neg B\neg C\neg D \vee \neg AB\neg CD \vee ABCD$; submodels : $|\mathrm{M_{CC}}(\Delta_5)| = 3,069$. (f) $\mathbf{m}(\Delta_6) : ABD \vee \neg ACD \vee \neg B\neg C\neg D \vee \neg ABC \vee \neg A\neg C\neg D$; submodels : $|\mathrm{M_{CC}}(\Delta_6)| = 1,102$.

method's developers, the present impasse in the debate on the utility of QCA is to a not inconsiderable degree due to the still-held belief among some prominent proponents of this method that common standards of methodological property evaluation for procedures of causal inference are not applicable to QCA.

Against this backdrop, it has been our objective to begin to put evaluations of QCA on a sound footing. This target has had three aspects: first, the establishment of a theoretical foundation for appropriate tests, including an explicit definition of configurational correctness as the decisive criterion; second, the justification for using inverse-search experiments with simulated data; and third, the design and execution of tests that respect the standards and search target of QCA.

We found that only the parsimonious search strategy performed correctly in all tests: under ideal conditions, when exposed to factor frame misspecification generating imperfect solution coverage and when confronted with limited empirical diversity. At the same time, we reiterate that these test designs do not represent all real-life recovery contexts. In particular, we did not test the performance of QCA when it is applied to data featuring imperfect consistencies or both limited diversity and factor frame misspecification with imperfect solution coverage. We will subject QCA to further tests in future work. Still, given that we have exposed all existing simulations as decidedly defective, no reasons for abandoning the method currently exist when researchers are working in recovery contexts corresponding to one of our test designs and they neither use conservative nor intermediate solutions.[30]

Notwithstanding these reassurances, the fact that only solutions generated via QCA's parsimonious search strategy seem guaranteed to be correct has major implications, not least because the intermediate search strategy that was introduced about a decade ago is increasingly favored by most methodologists and applied users of QCA. One such implication is that, if conservative and intermediate solutions frequently infer more than warranted, the most conservative QCA solution type in the literal sense of the word "conservative" is, ironically, the one of which it has so far been claimed that it is the most liberal. A radical rethink on some hitherto central tenets of QCA methodology is, therefore, urgently needed. Another major implication is that a significant share of results presented in scientific publications that have made use of a search strategy other than the parsimonious one are most probably not supported by those empirical data assumed to have generated them. In consequence, researchers using QCA for causal inference, particularly in human-sensitive areas such as public health and medicine, should immediately discontinue employing the method's conservative and intermediate search strategy.

Authors' Note

Both authors contributed equally to this work.

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Supplemental Material

The online [appendices/data supplements/etc.] are available at http://journals.sagepub.com/doi/suppl/10.1177/0049124117701487.

Notes

1. Some components of Qualitative Comparative Analysis (QCA) are also used for purposes other than causal data analysis. For instance, Mendel and Korjani (2012) employ fuzzy-set minimization techniques for linguistic summarization. The conclusions of the present article only concern applications of QCA in contexts of causal inference.
2. This finding does not imply that certain second-order procedures involving the parsimonious solution type such as Schneider and Wagemann's (theory guided)

enhanced standard analysis are also correct (Cooper and Glaesser 2016; Thiem 2016c).

3. Some textbooks use the term soundness as a synonym for correctness (e.g., Boolos, Burgess, and Jeffrey 2007:ch. 14).
4. For instance, regressional analytic methods impose the Gauss–Markov assumptions (Gelman and Hill 2007:45-47), and Bayesian network methods rely on the Causal Markov and Faithfulness assumptions (Spirtes, Glymour, and Scheines 2000:29-31).
5. Ragin and Rihoux (2004:10) and Wagemann and Schneider (2010:378) suggest that QCA does not require any background assumptions but do not provide any reasons. However, all procedures of causal inference come with a set of background assumptions as it is impossible to infer causation without assumptions about the unmeasured causal background (Cartwright 1989: 55-90).
6. In the fourth section, we discuss an application of QCA that results in correct models although configurational homogeneity is violated.
7. Data-driven tests are not the only approach to correctness evaluation. Certain algorithms can also be shown to be correct by mathematical proof.
8. Emphasis in quote added.
9. Rohlfing (2016) puts forward some ideas for consideration with regard to the design of adequate simulations but rejects the argument that simulations based on a single data set could yield any generally meaningful insights (2016:2075). We disagree. If only a single, properly designed and adequately implemented simulation shows QCA to draw incorrect causal inferences under perfectly ideal conditions, the method will have to be judged incorrect. What cannot be inferred from a single simulation, however, is the conclusion that QCA is correct.
10. Seawright (2014:118-19) is the only exception. With respect to Lucas and Szatrowski (2014), he argues that their "results provide the valuable insight that QCA fails, often quite badly, when asked to analyze data that are generated through some process other than Boolean truth functions. However, QCA supporters may complain with some justification that the simulations simply do not take their ontological commitments seriously."
11. The reason is the following: In δ_{LS}, for each minterm Ψ_i which differs in exactly one position from another minterm Ψ_j, it holds that Ψ_i is combined with $Y = 0$ whereas Ψ_j is combined with $Y = 1$. Fiss, Marx, and Rihoux (2014:96-97) also miss this peculiarity in Lucas and Szatrowski's (2014) data when they argue that "the complex and intermediate solutions are not valid tests given that these are synthetic data that do not cover all possible configurations."
12. Ragin (2014:92-93), Fiss et al. (2014:97), and Vaisey (2014:108) also note this error.

13. Correlations and implications are logically unrelated dependencies. The former hold among variables, the latter among values of variables. Moreover, two variables can be correlated without any of their values being implicationally dependent and vice versa.
14. In the fourth section, we show how to ensure implicational independence of a factor in a more general way without the simulation of particular data.
15. Additional problems in Hug (2013) and Krogslund, Choi, and Poertner (2015) are exposed by Thiem (2014a) and Rohlfing (2015), respectively.
16. For instance, imagine a macrosociological study showed patriarchal statism, working-class mobilization, and a Catholic government to be causally relevant to social security program consolidation. Then, these results would be fully compatible with those of a subsequent study that had recourse to better data, such as the one by Hicks, Misra, and Ng (1995:340), and presented a model showing the combined presence of patriarchal statism, working-class mobilization, and a Catholic government, and the absence of unitary democracy to be causally relevant to social security program consolidation.
17. The QCApro package is a successor to the QCA package, with version 1.1-4 as its basis (Thiem and Duşa 2013b, 2013c).
18. The output of the full model space is important in the context of causal inference with configurational data, as opposed to contexts of noncausal applications of Boolean minimization such as logic synthesis, because the data-generating structure may well be among the models not returned (for details, see Baumgartner and Thiem 2015b).
19. A truth table is called saturated when each of its logically possible minterms has been instantiated at least as often as specified by the discretionary frequency cutoff—the lowest acceptable number of empirical instances of a minterm.
20. Italics removed from quote.
21. When there is no limited diversity, as in our previous tests, there is no need to distinguish between different search strategies.
22. An exception is Baumgartner (2015) who argues, on theoretical grounds, that only the parsimonious search strategy can systematically generate models that are amenable to a causal interpretation in accordance with the theory of causation implemented by QCA.
23. More precisely, whether a directional expectation regarding some factor is in agreement with a causal structure is based on counts of the occurrences of one factor level and its negation within the presupposed structure. If both levels occur equally often, no expectation is made. Other criteria for determining directional expectations can be conceived for evaluating intermediate solutions, but ours represent baseline rules that should provide the widest possible corridor of test results. We invite readers to experiment with other rules but predict with some confidence that our general conclusions will remain unchanged.

24. The diversity index sets the number of observed minterms in relation to the number of logically possible minterms (Boswell and Brown 1999:159-60).
25. By the requirements of **CC**, the return of two or more elements of $\mathbf{M}_{\mathbf{CC}}(\Delta')$ also leads to the test being passed. For example, it would be theoretically possible that models $\mathbf{m}_9$ and $\mathbf{m}_{10}$ in panels (i) and (j), respectively, of Figure 3 were returned as parts of the same solution (but not models $\mathbf{m}_9$ and $\mathbf{m}_{18}$ in panel (r), for instance, because if the former represented a redundancy-free model, the latter would not).
26. Exceptions with respect to the diversity index include Cress and Snow (1996), for example, whose truth table has a diversity index value of only 0.07 percent, and Smilde (2005), who analyses a truth table with 100 percent diversity.
27. Trivial truth tables are those with only negative or only positive output function values. For solutions with no model ambiguities, the only model is necessarily selected.
28. The number of correctness-preserving submodels for $\mathbf{m}(\Delta_1)$ to $\mathbf{m}(\Delta_6)$ varies considerably, but it is always higher than for $\mathbf{m}(\Delta')$. While $\mathbf{m}(\Delta_2)$ in panel (b) of Figure 5 generates only 33 submodels, no fewer than 3,069 submodels are produced by model $\mathbf{m}(\Delta_5)$ in panel (e).
29. Ragin (2014:92-93) has suspected that parsimonious solutions are “likely but not guaranteed” to pass the tests devised by Lucas and Szatrowski (2014). Our results indicate that they are even guaranteed to pass these tests.
30. Strictly speaking, our results are only applicable to crisp-set QCA but do not necessarily carry over to fuzzy-set QCA (Ragin 2008), multivalue QCA (Cronqvist and Berg-Schlosser 2009; Thiem 2013, 2015a), generalized-set QCA (Thiem 2014c), or QCA when applied to causal chains (Thiem 2015b). Also, formal evaluations still need to be performed for Coincidence Analysis (Baumgartner 2009, 2013; Baumgartner and Epple 2014; Baumgartner and Thiem 2015a), which is a configurational method closely related to QCA.

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